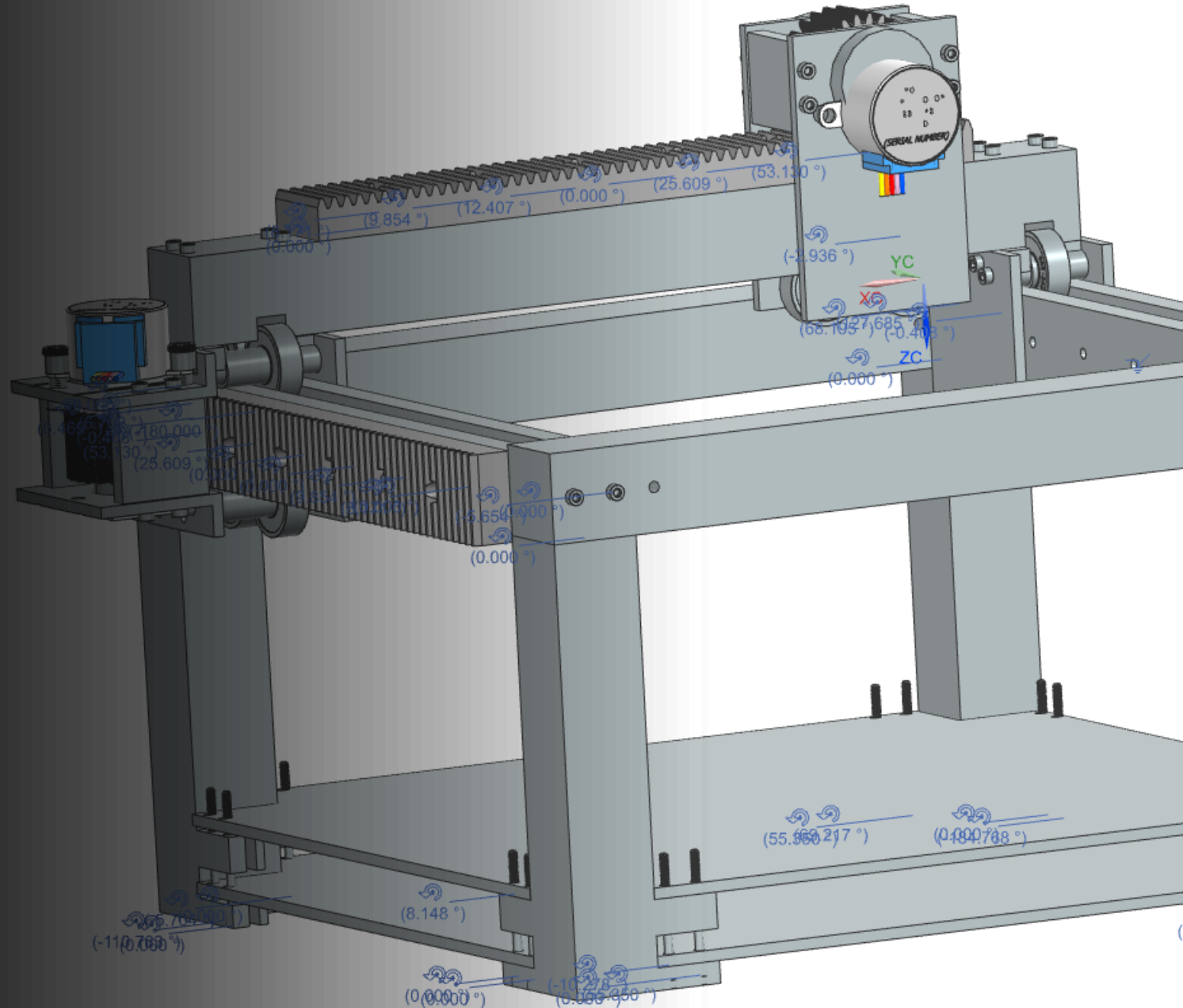


Order of the rings DP-2

Team 17: Elijah Freese,
Mariz Kedis, Natalie McDonald



Users struggle with organizing jewelry efficiently

Time we all like getting time back
the goal of our product is to give users back time by removing the step of organizing their jewelry and allowing them to get to other tasks



Our design is driven by clear user needs

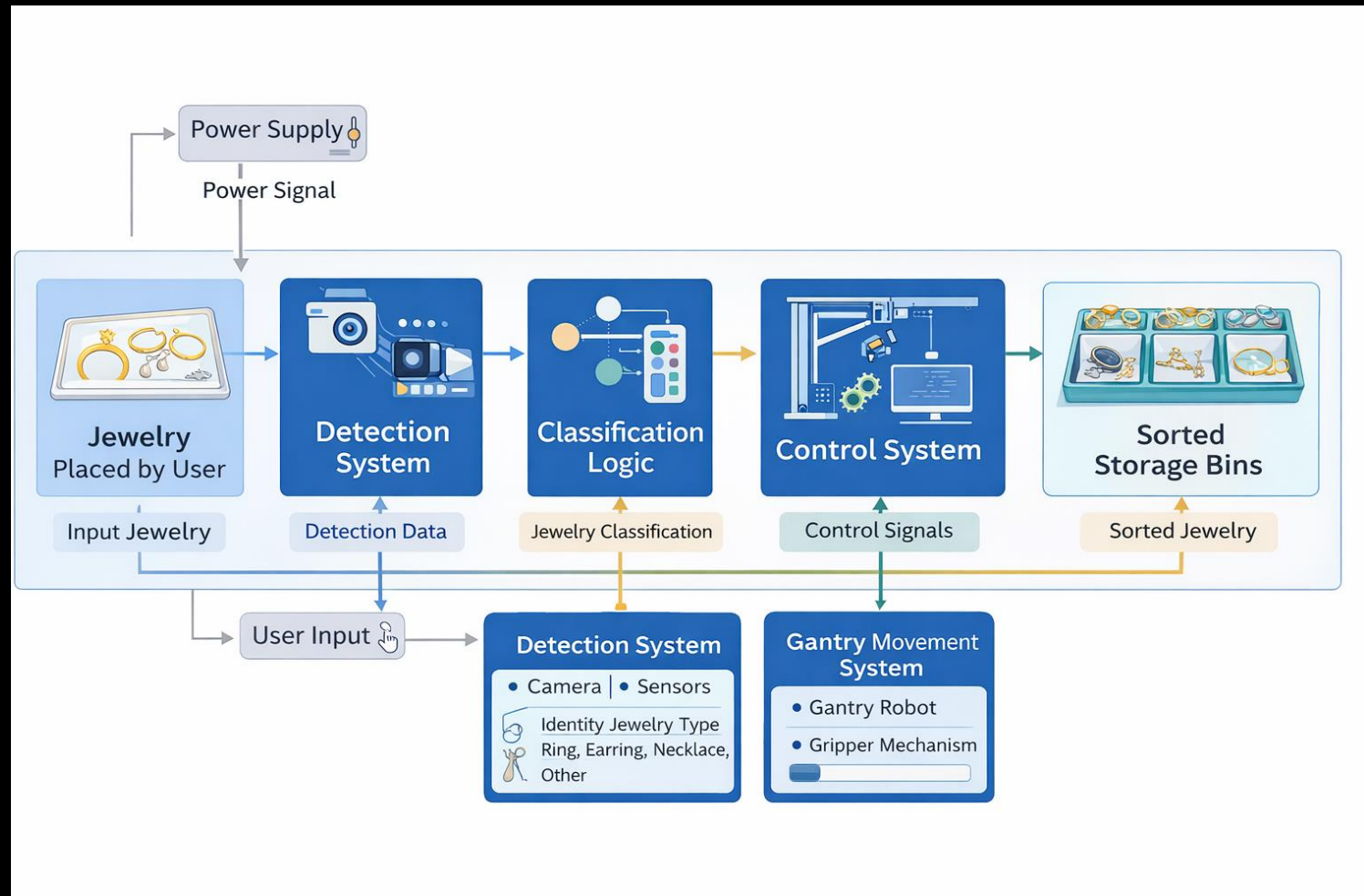
Mission - make a low-cost product that can save users time by automatically sorting jewelry .

Target user – any women young or old working or not.

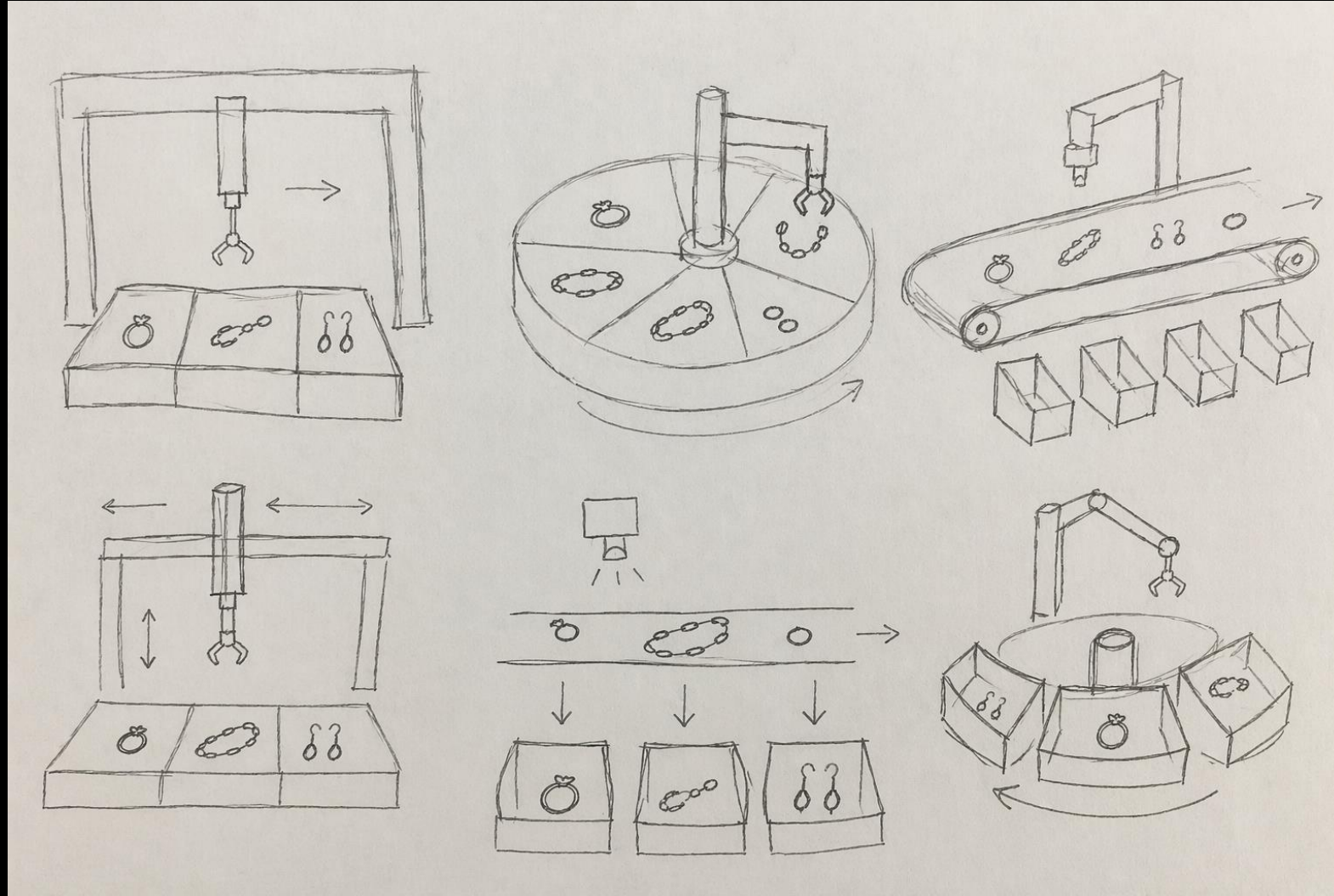
Specs – sorts 3 types of jewelry, each piece takes 30 seconds, is quiet, is serviceable, and is cost effective.

Customer needs – low cost, quick, small, and quiet.

The system
was broken
into key
functions



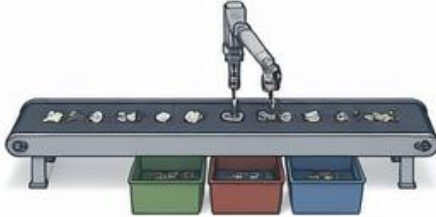
Multiple concepts were generated using structured ideation methods



Concepts were screened to eliminate weaker designs

CONCEPT SCREENING

Concepts were screened to eliminate weaker designs

SCREENING CRITERIA	1. GANTPY SYSTEM	2. CONVEYOR SYSTEM	3. ROTATING TRAY SYSTEM
 Feasibility Can it be built with available technology?			
 Cost Is it within the project budget?	✓	✓	✓
 Complexity Is it simple to design, build, and operate?	✓	✗	✓
	✓	✗	✗
KEEP?	KEEP	ELIMINATE	ELIMINATE

A weighted scoring method identified the best design

CONCEPT SCORING & SELECTION

A weighted scoring method identified the best design

EVALUATION CRITERIA	WEIGHT (%)	CONCEPTS		
		1. GANTRY SYSTEM	2. CONVEYOR SYSTEM	3. ROTATING TRAY SYSTEM
 Cost Lower cost is better	30%	8	5	6
 Simplicity Simpler to build and operate	30%	9	6	5
 Accuracy & Reliability Sorts items accurately and consistently	40%	9	6	7
TOTAL WEIGHTED SCORE	100%	8.8	5.8	6.2

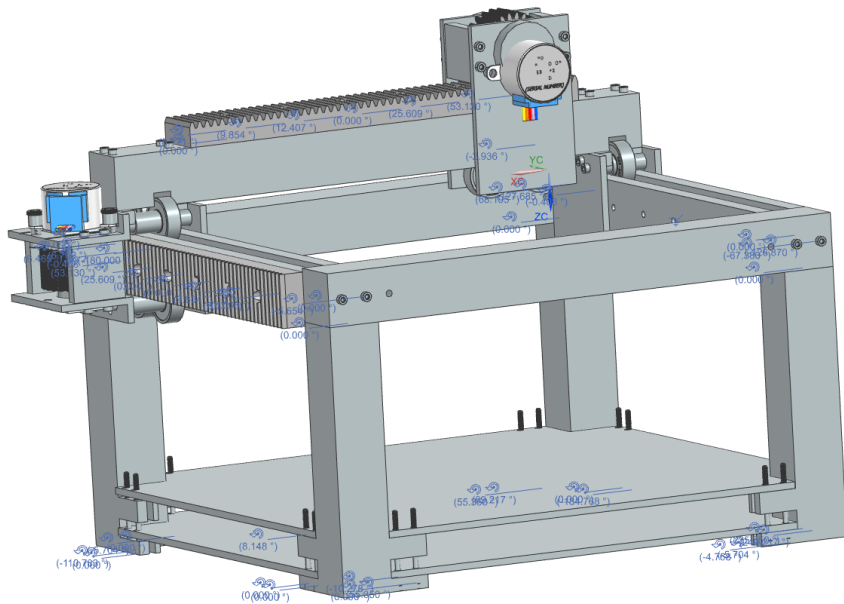


SELECTED CONCEPT: 1. GANTRY SYSTEM

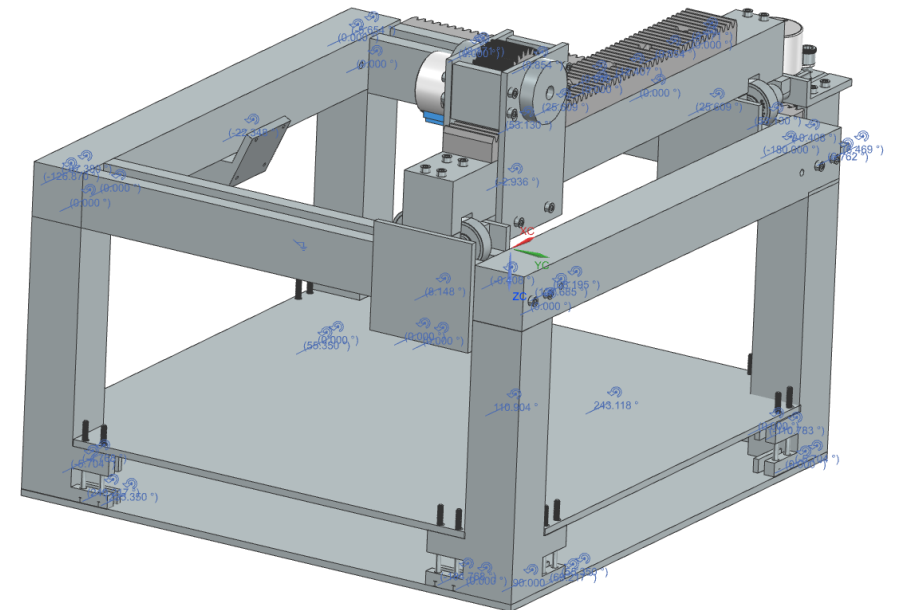
Highest overall score based on cost, simplicity, and accuracy.

User feedback led to key design improvements

Before



After



The final design efficiently sorts jewelry using an automated gantry system

